

RAGHAV ENGINEERING

Job Card — RRE-2026-0019

Basic Information			
Job Card No.	RRE-2026-0019	Received Date	11 May 2025
Due Date	28 May 2025	Customer	KSB — Seed Demo 18045
NBDN No.	TN25/0002/8510	PO No.	4507514470
Drawing No.	U6129/22/4	Heat No.	9752
Part No.	05439662	Quantity	5
Process	Welding, Machining	Welding Process	SMAW
Ring	FIIV		
Description	20 900 WCB Body — Overlay + Machining		

Customer / Material Details			
Product Group	CBE	Buyer	P. Gunasekaran
Material Code	01N0	Valve Size/Class	6" 900#
Valve Type/Comp.	Body	Base Material	WCB
Overlay Material	E8018-B2	Base Mat. Grade	E8018-B2
Regularization	WPS/RE/301	Ring Heat No.	A274
MPI / RT No.	M000216	Punching Details	RE/FIIV/DPOR/SROR
Other Details	RAG7208/25-26		

Operations / Routing								
#	Operation	Machine	Operator	Planned	Completed	Rejected	Status	
1	Pre-Machining	CNC-VTL-01	K. Selvam	5	5	0	Completed	
2	Welding	WLD-01	Gopi Krishnan	5	5	0	Completed	
3	Final Machining	CNC-VTL-01	K. Selvam	5	5	0	Completed	
4	Milling	MILL-01	K. Selvam	5	5	0	Completed	
5	Slitting	SLOT-01	K. Selvam	5	5	0	Completed	
6	Deburring	CNC-VTL-01	K. Selvam	5	5	0	Completed	

Welding Details			
Welding — Gopi Krishnan (ID: W-001)			
Weld Date	19 May 2025	Weld Metal	E8018-B2
Weld Height	4	Consumable Batch	D042934547
Consumable Brand	D&H Secheron	Product Name	Chromet 8018-B2
AWS Class	E8018-B2	Size	4.0mm
Mfg. Date	07 Apr 2025	Expiry Date	06 Apr 2027

Parameter	As per WPS	Actual	
Weld Qty	5	5	
Pre-heat (°C)	150	145	
Inter-pass (°C)	300	297	
Post-heat (°C)			
Amp	140-180	170	
Volt	24-30	26	
Travel Speed	130	142	
Gas Flow Rate			
Feed Rate			
Polarity	DCRP	DCRP	

PWHT Details			
Chart No.	A816	Process	Stress Relieving
Loading Temp	170	Loading Time	21

Soaking Temp	700	Soaking Time	240
Unloading Temp	175	Unloading Time	9

Air Testing & Inspection

Tester Name	R. Nagaraj	Pressure	6 bar
Duration	10 min	Result	pass

NDE / LPT — LPT

NDE Number	003 / Rev.07	Report No.	WPF/909
Test Coupon No.	TC-14	Deposit Thickness	14mm
Hardness Req.	OK	Duration	10 Min
Observer	S. Kumar	Inspected By	R. Nagaraj
Result	accepted		

Type	Chemical	Manufacturer	Batch No.	Expiry	
penetrant	24K32	Magnaflux	24K32	2029-10-01	
cleaner	22K01	Magnaflux	22K01	2029-10-01	
developer	22K16	Magnaflux	22K16	2029-10-01	

Machining / Dimensions

Machine Name	CNC-VTL-01	Operator	K. Selvam
Drawing Size	A2	Deposit Thk. Before	14mm
Deposit Thk. After	N/A		

Dimension	Required	Tolerance	Actual 1	Actual 2	Actual 3	P/F	
Milling — GSM	—	±0.1	N/A			na	
Milling — PSM	—	±0.1	N/A			na	
Milling — OAL	—	±0.1	N/A			na	
Milling — OD	—	±0.1	N/A			na	
Milling — ID	—	±0.1	N/A			na	
Milling — Top / OAH	—	±0.1	N/A			na	
Soft / Pre Machining — GSM	—	±0.1	N/A			na	
Soft / Pre Machining — PSM	—	±0.1	N/A			na	
Soft / Pre Machining — OAL	—	±0.1	N/A			na	
Soft / Pre Machining — OD	—	±0.1	N/A			na	
Soft / Pre Machining — ID	—	±0.1	N/A			na	
Soft / Pre Machining — Top / OAH	—	±0.1	N/A			na	
Hard Facing — GSM	—	±0.1	N/A			na	
Hard Facing — PSM	—	±0.1	N/A			na	
Hard Facing — OAL	—	±0.1	N/A			na	
Hard Facing — OD	—	±0.1	N/A			na	
Hard Facing — ID	—	±0.1	N/A			na	
Hard Facing — Top / OAH	—	±0.1	N/A			na	
Bore Diameter	150.00	±0.05	150.02	150.01	150.03	pass	

Dispatch

DC Number	DC-2025-220199	Dispatch Date	30 May 2025
Vehicle Details	TN-01-AB-1234		

Approvals

Gopi Krishnan Production (19 May 2025)	R. Nagaraj Quality Control (20 May 2025)	S. Muthu Stores (21 May 2025)
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